# 🛡️ Module 08: Windows Exploitation and Privilege Escalation

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# Exercise 01: Enumerating SMB
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### Scenario

The Server Message Block (SMB) protocol is a critical target during Windows internal network assessments due to its widespread usage for file sharing, printer access, and remote administration. Misconfigured SMB services often leak domain infrastructure details, OS version information, NetBIOS Computer Names, and Active Directory domain metadata without requiring prior authentication.

Security analysts use Nmap Scripting Engine (NSE) scripts—such as **smb-os-discovery** (**nmap --script smb-os-discovery 192.168.0.7**) and default script suites (**nmap -sC 192.168.0.7**)—to enumerate target hosts. Executing SMB OS discovery against target server **192.168.0.7** retrieves target system metadata, including its DNS Domain Name (**webserver**), NetBIOS computer name, and supported SMB dialect versions.

In this exercise, we initialize Nmap scanning on Parrot Security, run SMB OS discovery scripts against target server **192.168.0.7**, examine default SMB enumeration script outputs, analyze debug traces using the `-d` switch, and extract the target's DNS domain name (**webserver**).

### Objective

The objective of this lab is to help students learn how to:
- Identify active SMB services running on Windows targets (**192.168.0.7**)
- Execute targeted Nmap SMB OS discovery scripts (**nmap --script smb-os-discovery 192.168.0.7**)
- Perform comprehensive default script scanning (**nmap -sC 192.168.0.7**) to enumerate SMB shares and security modes
- Analyze Nmap debug output traces (**nmap -d -sC 192.168.0.7**) to inspect SMB login sequence responses
- Extract critical network metadata, specifically identifying the target's DNS Domain Name (**webserver**)

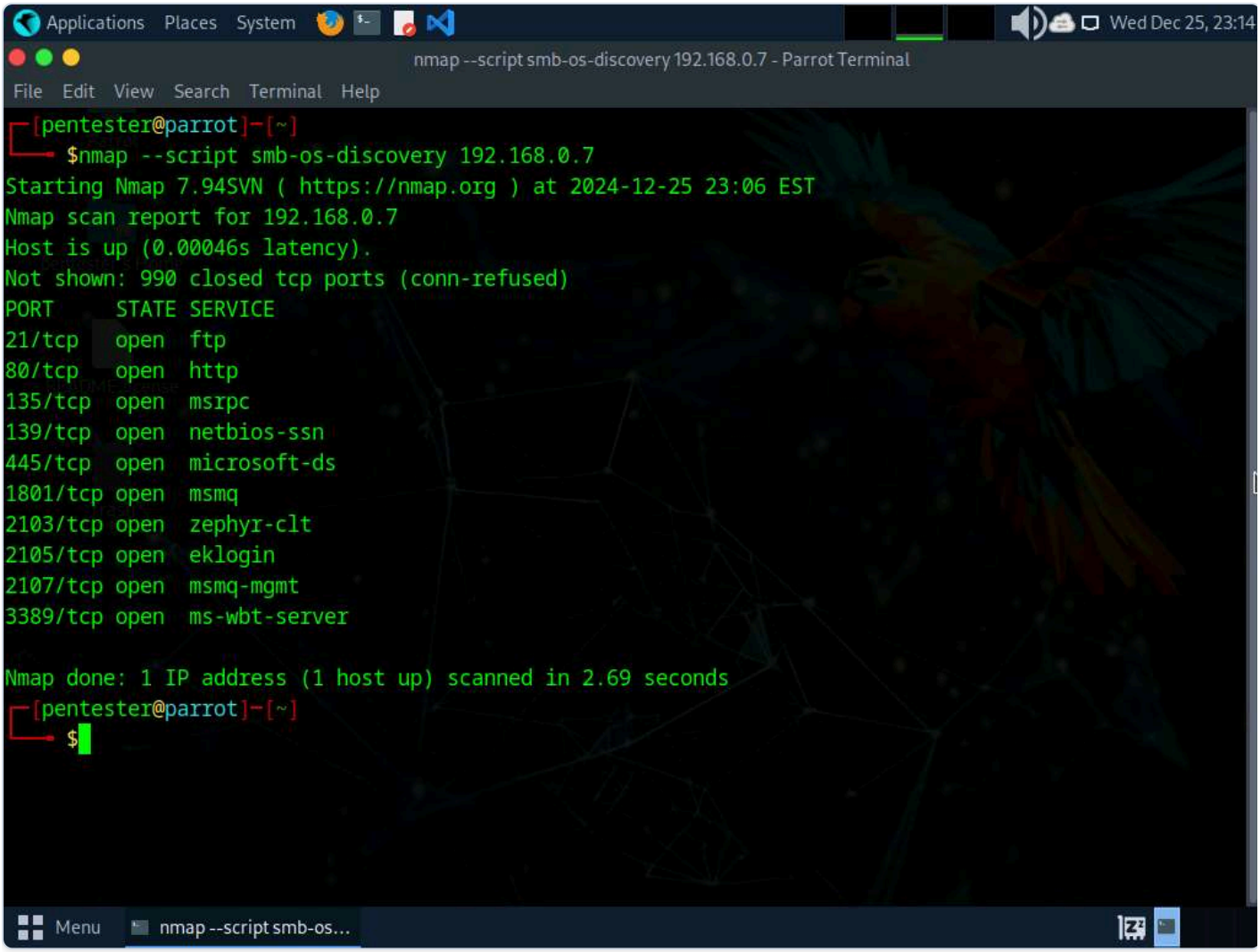
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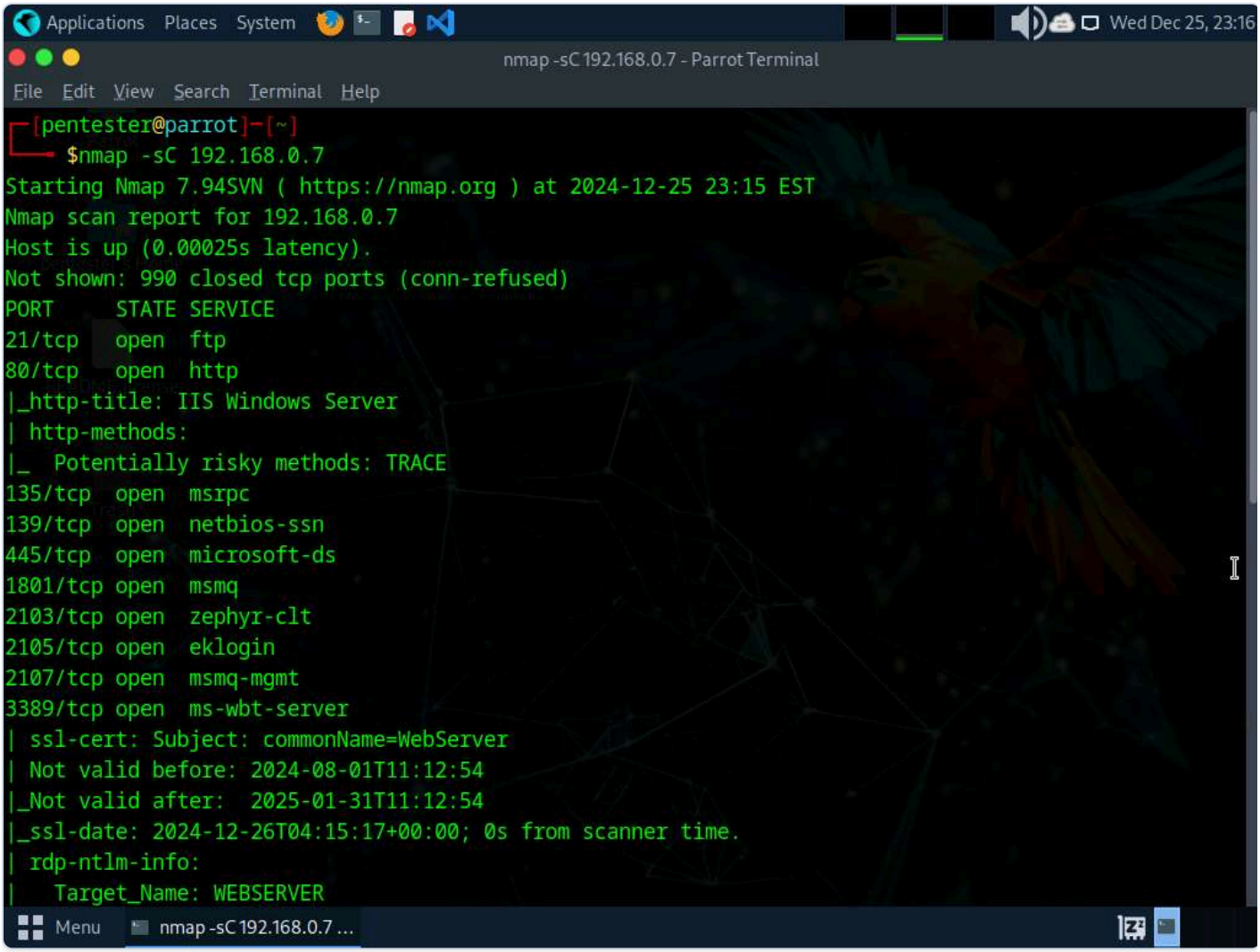
## Task 1: Perform SMB Protocol Enumeration and OS Discovery

In this task, we launch terminal scans on Parrot Security, run `smb-os-discovery` against **192.168.0.7**, execute default Nmap scripts, and analyze debug traces to discover system domain parameters.

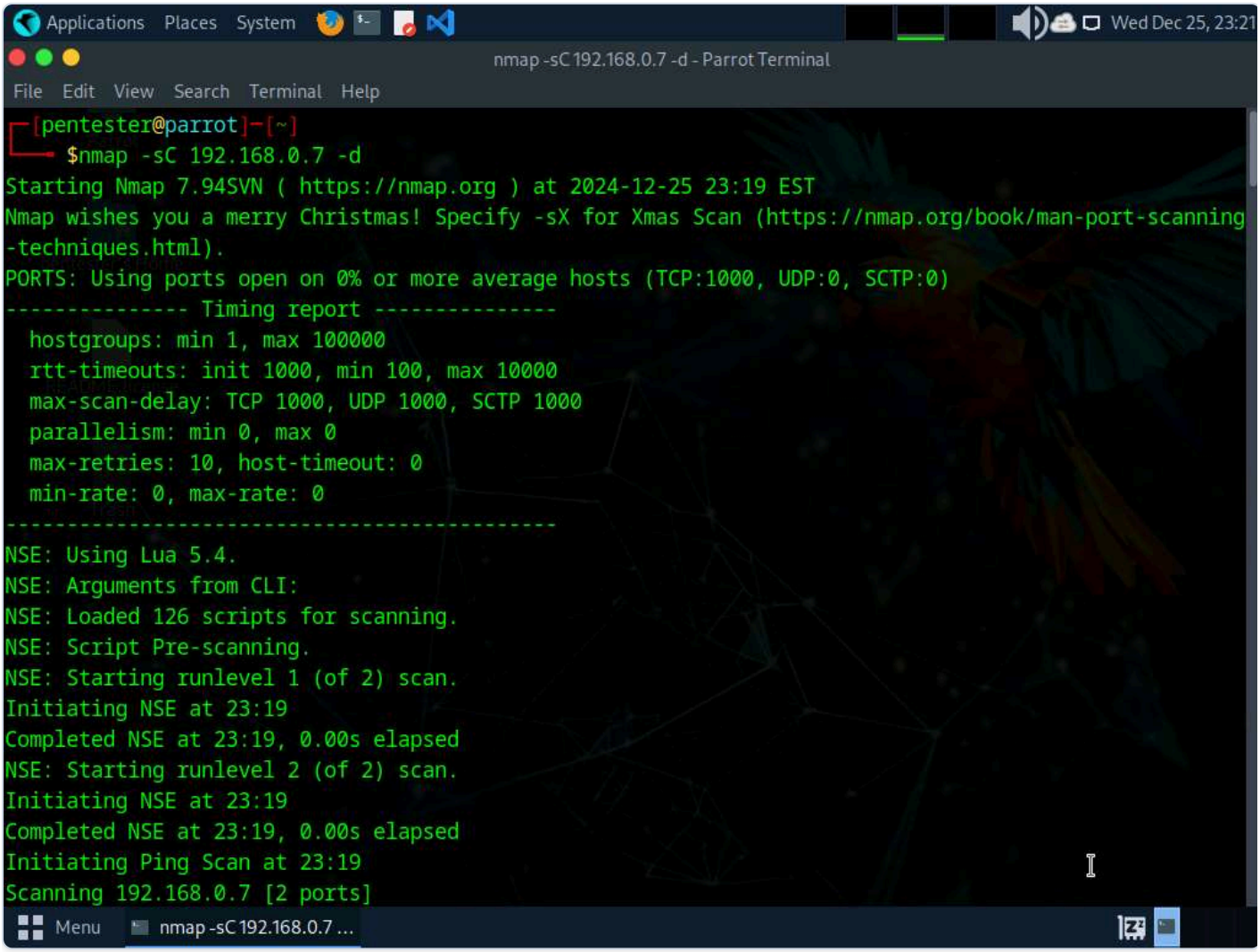
### Lab Tasks

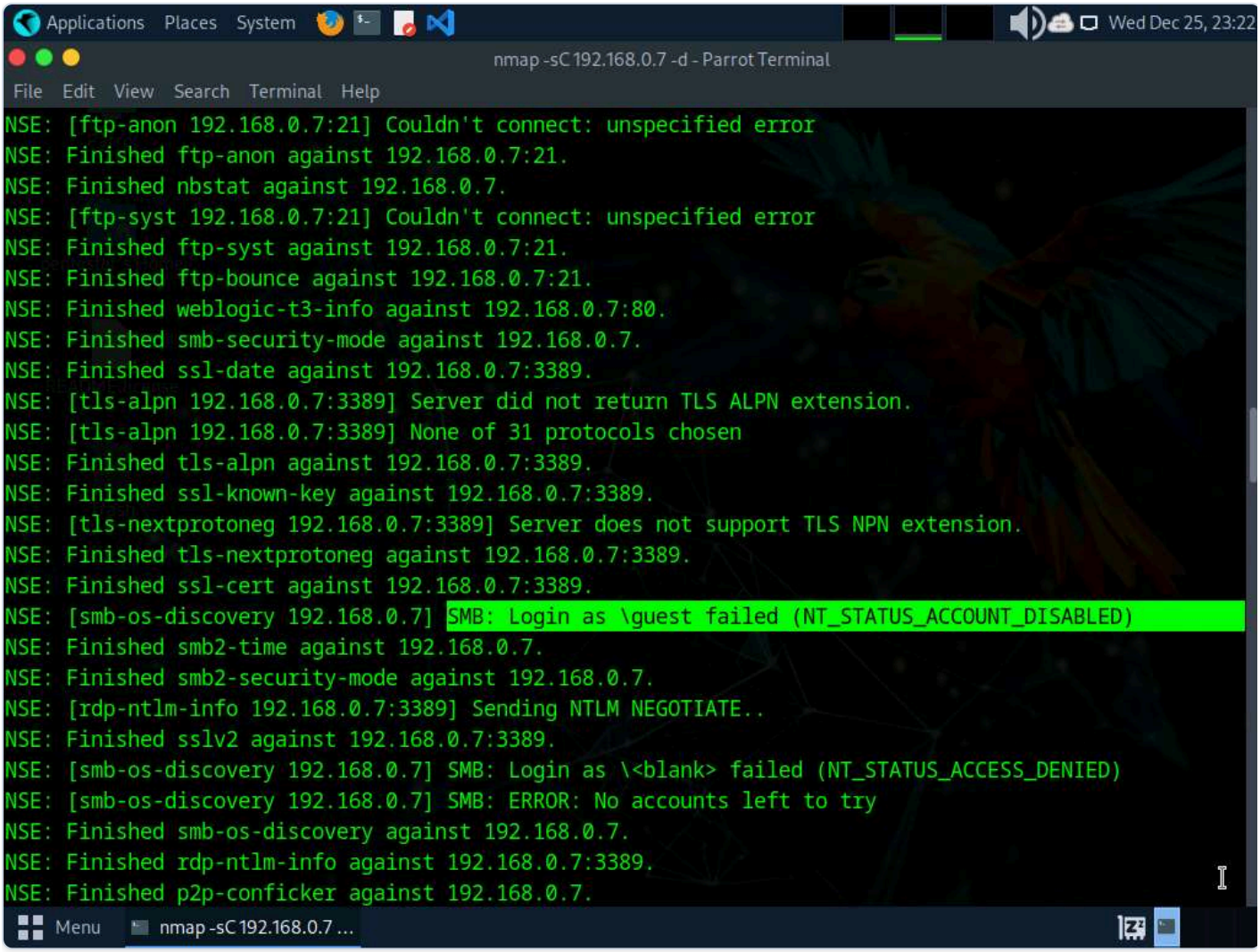
1. Switch to **Parrot Security** (**toor**). Open a terminal window.
2. Run Nmap SMB OS discovery script against the target IP address:
   **nmap --script smb-os-discovery 192.168.0.7**
3. Perform default script enumeration to gather detailed SMB service details:
   **nmap -sC 192.168.0.7**
   *Observe the detailed output revealing OS version, NetBIOS Computer Name, Workgroup, and DNS Domain Name (**webserver**).*


Applications Places System
Wed Dec 25, 23:14
nmap --script smb-os-discovery 192.168.0.7 - Parrot Terminal
File Edit View Search Terminal Help
[pentester@parrot]-[~]
$nmap --script smb-os-discovery 192.168.0.7
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-25 23:06 EST
Nmap scan report for 192.168.0.7
Host is up (0.00046s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT STATE SERVICE
21/tcp open ftp
80/tcp open http
135/tcp open msrpc
139/tcp open netbios-ssn
445/tcp open microsoft-ds
1801/tcp open msmq
2103/tcp open zephyr-clt
2105/tcp open eklogin
2107/tcp open msmq-mgmt
3389/tcp open ms-wbt-server
Nmap done: 1 IP address (1 host up) scanned in 2.69 seconds
[pentester@parrot]-[~]
$
Menu
nmap --script smb-os...




1. Run the scan with debug logging enabled to analyze protocol-level interactions and authentication trace attempts:
   **nmap -d -sC 192.168.0.7**


Applications Places System
Wed Dec 25, 23:21
nmap -sC 192.168.0.7 -d - Parrot Terminal
File Edit View Search Terminal Help
[pentester@parrot]-[~]
$nmap -sC 192.168.0.7 -d
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-25 23:19 EST
Nmap wishes you a merry Christmas! Specify -sX for Xmas Scan (https://nmap.org/book/man-port-scanning-techniques.html).
PORTS: Using ports open on 0% or more average hosts (TCP:1000, UDP:0, SCTP:0)
---------- Timing report ----------
hostgroups: min 1, max 100000
rtt-timeouts: init 1000, min 100, max 10000
max-scan-delay: TCP 1000, UDP 1000, SCTP 1000
parallelism: min 0, max 0
max-retries: 10, host-timeout: 0
min-rate: 0, max-rate: 0
----------
NSE: Using Lua 5.4.
NSE: Arguments from CLI:
NSE: Loaded 126 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 23:19
Completed NSE at 23:19, 0.00s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 23:19
Completed NSE at 23:19, 0.00s elapsed
Initiating Ping Scan at 23:19
Scanning 192.168.0.7 [2 ports]
Menu
nmap -sC 192.168.0.7 ...




1. Inspect error logs confirming failed guest login attempts typical of modern hardened Windows hosts. Close the terminal window.

This concludes the lab exercise.

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> **IMPORTANT:** ❓ **Question 8.1.1.1** Utilize the Parrot Security machine to conduct a scan of the 192.168.0.7 system, identifying the SMB service and enumerating information from the SMB protocol. Retrieve and provide the DNS_domain_name associated with the 192.168.0.7 machine. [!SUCCESS] ✅ **Answer:** **webserver**